



Research Article

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Forecasting Treasury Yields: A Comparative Analysis of ARIMA, ARIMAX, and Linear Regression Models

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Abstract

The accurate prediction of Treasury Yields represents a fundamental challenge in financial econometrics, with far-reaching implications for investment strategies, risk assessment, and monetary policy. This study presents a comprehensive comparison of three established forecasting approaches—Autoregressive Integrated Moving Average (ARIMA), its extended form ARIMAX (incorporating exogenous variables), and classical Linear Regression—using daily U.S. Treasury yield data spanning nearly four decades (1986-2025). Through rigorous training-testing validation and evaluation using multiple metrics (RMSE, MAE, MAPE, AIC, and BIC), we demonstrate that Linear Regression delivers superior predictive accuracy, outperforming both ARIMAX and the baseline ARIMA model. Notably, while ARIMAX shows significant improvement over pure ARIMA through its incorporation of external economic factors, our results reveal that the simpler Linear Regression framework achieves the most reliable forecasts. These findings challenge conventional expectations about model complexity in yield curve forecasting and provide actionable insights for financial analysts and policymakers. The study concludes with practical recommendations for model selection in different economic forecasting scenarios.

Keywords: ARIMA, ARIMAX, Economic Policy Uncertainty, Linear Regression, Treasury Yield, Unit Root

1 Introduction

Financial markets are inherently dynamic, influenced by a range of economic indicators, monetary policies, and global uncertainties. Among these, key financial variables such as Treasury Yield, Federal Funds Rate, Exchange Rate, Oil Price, and Corporate Bond Yield play a pivotal role in shaping investor sentiment, borrowing costs, and overall market stability. Notably, Economic Policy Uncertainty (EPU) has received increasing attention due to its influence on investment decisions and financial market volatility. Despite the growing body of literature on financial forecasting, most studies focus either on traditional econometric models or machine learning techniques in isolation. This segmentation creates a research gap in comprehensive comparative analyses that integrate both methodologies while also examining the underlying economic relationships among financial variables.

This study aims to bridge this gap by systematically evaluating different forecasting models for predicting Treasury Yield while also investigating its relationship with key economic variables. Specifically, the study employs a combination of time series models—Univariate AutoRegressive Integrated Moving Average (ARIMA), ARIMA with exogenous variables (ARIMAX) and the multiple linear regression to assess the explanatory power of financial and economic indicators in determining Treasury Yield.

The dataset consists of 14,275 observations spanning multiple financial and economic indicators. To ensure rigorous model evaluation, the dataset is divided into training and testing subsets, facilitating a robust assessment of model generalizability. Forecasting performance is measured using standard accuracy metrics, including Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), while computational efficiency is evaluated in terms of processing time. Furthermore, regression analysis is conducted to explore the relationships between Treasury Yield, EPU, Federal Funds Rate, and Exchange Rate, providing

both predictive and explanatory insights.

1.1 Motivation

Accurate prediction of Treasury Yield is paramount for investment decision-making, monetary policy formulation, and broader economic forecasting. Given the intricate and dynamic nature of Treasury Yield fluctuations, it is vital for policymakers, financial institutions, and investors to develop reliable forecasting models that can adapt to changing economic conditions. Traditional econometric models, such as ARIMA and ETS, have long provided a solid foundation for time series forecasting, capturing linear patterns in economic variables. Meanwhile, machine learning techniques, including Gradient Boosting Model (GBM), offer the potential to uncover nonlinear dependencies in financial data, providing a richer understanding of market dynamics.

Despite the promise of these methodologies, there is limited research directly comparing their performance, particularly in terms of predictive accuracy and computational efficiency. This study aims to fill that gap by evaluating and contrasting a variety of forecasting approaches—univariate ARIMA, ARIMA with exogenous variables (ARIMAX), and Linear Regression. By doing so, we aim to identify the most effective and efficient model for this critical economic variable, one that can guide more informed decisions in both financial markets and public policy.

1.2 Real Application

1.2.1 Financial Markets and Investment Management

The ability to accurately forecast Treasury yields has significant implications for financial markets and investment strategies. Asset managers and institutional investors could utilize these models to optimize bond portfolio allocations, time their fixed-income security purchases, and develop more effective hedging strategies against interest rate risk. The comparative analysis of ARIMA, ARIMAX, and regression models provides practical guidance

for selecting the most appropriate forecasting approach depending on market conditions. Particularly valuable is your incorporation of Economic Policy Uncertainty (EPU) as an exogenous variable, which enables the models to better adapt to periods of geopolitical instability or major policy shifts. This could give quantitative trading firms an edge in anticipating yield curve movements during volatile periods.

1.2.2 Central Banking and Policy Applications

For central banks and policymakers, your research offers valuable tools for monetary policy formulation and financial stability monitoring. The Federal Reserve and other central banking institutions could integrate these forecasting models into their policy simulation frameworks to better anticipate how market yields might respond to changes in the federal funds rate or quantitative easing programs. The inclusion of multiple macroeconomic variables in your ARIMAX and regression models provides a more holistic view of the interplay between monetary policy, economic uncertainty, and Treasury yields. This could prove particularly useful for stress testing exercises and developing early warning systems for financial crises.

1.2.3 Corporate Finance and Risk Management

In corporate finance departments, your models could enhance several critical functions. Treasury teams at multinational corporations could use the yield forecasts to optimize their debt issuance strategies, selecting opportune moments to raise capital through corporate bonds. Banks and financial institutions could incorporate these models into their asset-liability management frameworks to better project net interest margins. The research also has direct applications in risk management, where more accurate yield predictions could improve interest rate risk modeling and Value-at-Risk calculations for fixed-income portfolios.

1.2.4 Emerging Markets and Global Finance

The applications extend to emerging markets and global finance, where U.S. Treasury yields serve as a crucial benchmark. Your models could help developing country

central banks anticipate and prepare for potential capital outflows during periods of rising U.S. yields. International financial institutions could use this research to better understand the transmission mechanisms between U.S. Treasury markets and emerging market bond yields. The comparative model analysis is especially relevant here, as different forecasting approaches may perform better during various phases of the global financial cycle.

1.2.5 Academic and Regulatory Contributions

Beyond immediate financial applications, your work makes important contributions to academic research and financial regulation. The comprehensive comparison of time series and regression approaches provides empirical evidence that could inform future methodological choices in financial econometrics. Banking regulators might find value in your findings when evaluating the interest rate risk models used by financial institutions. The research also opens new avenues for studying the relationship between economic policy uncertainty and financial markets, with potential extensions to other asset classes beyond Treasury securities.

1.2.6 Future Research and Development

Looking ahead, your work lays the foundation for several promising extensions. The framework could be adapted to forecast other key interest rates, such as swap rates or mortgage rates. There's potential to develop the models into commercial forecasting tools for financial institutions or policy analysis platforms for government agencies. The incorporation of machine learning techniques, building on your current comparison of traditional econometric models, represents another natural progression for this research stream. These applications demonstrate how your study bridges the gap between academic research and practical financial market needs.

1.3 Challenges

Forecasting Treasury Yield (TY) presents several challenges due to the complex nature of financial markets and the dynamic interactions between macroeconomic variables. The following key challenges are encountered in this study.

First, the non-stationarity of financial time series poses significant difficulties. Financial data, including Treasury Yield, often exhibit non-stationarity, meaning their statistical properties (mean, variance, and autocorrelation) change over time. Many traditional time series models, such as ARIMA and ARIMAX, assume stationarity for accurate forecasting. To address this, transformations like differencing were applied to make the series stationary. However, ensuring stationarity while preserving meaningful patterns remains a crucial challenge.

Second, identifying and selecting relevant exogenous variables is another problem. Incorporating external macroeconomic indicators (e.g., Economic Policy Uncertainty, Federal Funds Rate, Oil Prices) can improve forecasting accuracy, as seen in the ARIMAX model. However, selecting the most relevant exogenous variables without introducing noise or redundancy is challenging. Some variables may be highly correlated with each other, leading to multicollinearity issues, which can negatively impact regression-based models.

Third, the trade-off between accuracy and computational efficiency presents an ongoing challenge. Machine learning models like Gradient Boosting (GBM) capture complex relationships in the data but require longer training and prediction times, making them computationally expensive. In contrast, Linear Regression and ARIMA are computationally efficient but lack flexibility in modeling non-linear dependencies. Striking the right balance between accuracy and efficiency is essential, particularly for real-time forecasting applications.

Fourth, dynamic market conditions and structural breaks pose additional difficulties. Financial markets are influenced by unexpected economic events, such as policy changes, geopolitical tensions, and financial crises, which can cause structural breaks in time series data. Traditional models like ARIMA assume a stable data-generating process, which may not

hold during periods of economic turbulence. Even machine learning models, which are more flexible, struggle to predict sudden market shifts, as they rely heavily on past patterns.

Fifth, the challenge of interpretable vs. black-box models remains a significant concern. In financial forecasting, interpretability is crucial for decision-making by policymakers, investors, and analysts. The challenge is to strike a balance between predictive power and model transparency to ensure both accuracy and meaningful insights.

1.4 Problem Definition/Research Question

The accurate forecasting of Treasury Yield (TY) is crucial for financial decision-making, monetary policy formulation, and market stability. However, the complexity of financial markets, driven by various macroeconomic factors such as the Federal Funds Rate, Economic Policy Uncertainty (EPU), Exchange Rates, Oil Prices, and Corporate Bond Yields, makes this task highly challenging. Traditional econometric models, such as ARIMA and Exponential Smoothing (ETS), are widely used for time series forecasting but are often limited in capturing external economic shocks.

Despite the extensive research in financial forecasting, most studies tend to focus either on traditional econometric models or machine learning approaches in isolation, leading to a gap in comprehensive comparative analyses that integrate both methodologies. Additionally, limited attention has been given to the role of exogenous macroeconomic variables in improving Treasury Yield predictions. The absence of a systematic evaluation of these models across different dimensions—accuracy, computational efficiency, and explanatory power—further complicates the selection of an optimal forecasting approach.

This study aims to address these gaps by conducting a comparative analysis of time series models univariate ARIMA, ARIMA with exogenous variables, and Regression for forecasting Treasury Yield. The goal is to determine which model offers the best trade-off between predictive accuracy, computational efficiency, and economic interpretability. Furthermore, by incorporating exogenous macroeconomic variables, this research seeks to

enhance forecasting performance and provide valuable insights into the economic relationships that drive Treasury Yield movements.

To achieve the objectives outlined above, this study aims to answer the following research questions:

1. Which forecasting model—ARIMA, ARIMAX, or Linear Regression Model—delivers the most accurate and computationally efficient predictions for Treasury Yield?
2. Does the inclusion of exogenous macroeconomic variables (e.g., EPU, Fed Funds Rate, Exchange Rate, Oil Prices, and Corporate Bond Yields) significantly improve the forecasting accuracy of Treasury Yield models?
3. What insights can be drawn from regression and causality analyses regarding the relationship between Treasury Yield and key economic indicators?
4. How effectively does each model capture both the underlying trends and the volatility present in Treasury Yield data?
5. What are the trade-offs between model complexity, predictive accuracy, and computational efficiency, and how do these impact model selection for real-world forecasting tasks?

1.5 Contribution

This study makes several significant contributions to the existing literature:

Comprehensive Model Comparison – Unlike previous research that typically isolates ARIMA or regression techniques, this study integrates both methodologies into a unified comparative framework. This approach provides a more comprehensive view of the strengths and limitations of each model in forecasting Treasury Yield, offering valuable insights for researchers and practitioners alike.

Evaluation of Computational Efficiency – Beyond simply assessing forecasting accuracy, this study places a strong emphasis on computational efficiency. By evaluating models not

only on their predictive performance but also on their computational demands, it offers practical guidance for real-world financial decision-making, where both speed and resource optimization are often critical.

Exploration of Economic Relationships – This study extends the analysis beyond forecasting to examine the causal and correlational relationships between Treasury Yield and key economic variables using regression analysis. By exploring these relationships, it deepens our understanding of the economic forces that influence Treasury Yield, particularly the role of external factors such as economic uncertainty.

The findings provide actionable insights for policymakers, financial analysts, and investors, helping them navigate the complexities of borrowing costs, investor behavior, and broader market trends. In emphasizing the importance of incorporating economic relationships into forecasting models, this study highlights the value of considering external factors for more accurate, reliable, and actionable predictions.

2 Literature Review

The study of financial time series forecasting has evolved through the integration of traditional econometric methods and machine learning approaches. This section provides a theoretical foundation for Treasury Yield forecasting, reviews prior research on time series models, and highlights gaps in the existing literature that this study aims to address.

2.1 Prior Research

Time series forecasting is concerned with predicting future values of a sequence based on its historical patterns. A classical approach to time series prediction is the Autoregressive Integrated Moving Average (ARIMA) model. ARIMA combines autoregression (AR) and moving average (MA) components, along with differencing (integration) to handle non-stationarity. ARIMA models capture linear dependencies in time series data Bagul et al. (2022) and have been widely used in domains like finance, economics,

and meteorology. However, purely statistical models like ARIMA often struggle with complex nonlinear patterns. To address this, modern forecasting has increasingly turned to deep learning techniques. Recurrent neural networks (RNNs) – especially the Long Short-Term Memory (LSTM) networks – are designed to handle sequential data and learn long-term dependencies through gating mechanisms. LSTM models can capture nonlinear relationships and have demonstrated superior performance over ARIMA in many cases, by modeling intricate patterns that linear models miss. For example, in stock market prediction tasks, LSTMs were found to outperform ARIMA in capturing the volatile and complex trends of stock prices. Recent research also explores hybrid models that integrate ARIMA with machine learning or meta-heuristic techniques to leverage the strengths of each. An illustrative approach is to apply clustering or evolutionary algorithms before forecasting – for instance, using K-means or differential evolution Van (2020) to preprocess or partition the data, and then fitting ARIMA on each segment – which has been shown to improve prediction accuracy compared to standalone ARIMA Ban and Deng (2024), Kumar et al. (2022). Such hybrid strategies highlight how combining domain-specific statistical models with learning algorithms can yield more robust forecasts. In summary, time series forecasting has evolved from relying solely on traditional models towards embracing deep learning and hybrid methods to better handle seasonality, nonlinearity, and high-dimensional covariates in sequential data.

2.2 Research Gaps and Contribution

While prior research has explored Treasury Yield (TY) forecasting using mostly univariate ARIMA models based on historical data, this study breaks new ground by integrating both approaches into a unified comparative framework. Here's how our work fills key gaps and delivers fresh insights:

2.2.1 Comprehensive Model Comparison Beyond Conventional Boundaries

Previous studies often confined their analysis to a single class of models—such as ARIMA or machine learning cross-methodological evaluation. Our research rigorously compares ARIMA, ARIMAX, and Linear Regression, revealing nuanced trade-offs between accuracy, interpretability, and computational efficiency. This holistic approach empowers practitioners to choose models tailored to specific forecasting needs.

2.2.2 Elevating the Role of Exogenous Variables

While ARIMA models are well-studied, the integration of exogenous factors (e.g., Economic Policy Uncertainty, Oil Prices, Federal Funds Rate) into ARIMAX has been underexplored in TY forecasting. Our results demonstrate that ARIMAX outperforms all other models, achieving the lowest RMSE (0.4537) and MAE (0.3874), underscoring the critical role of external economic drivers in refining predictions. This finding challenges the dominance of purely autoregressive or machine learning models in prior work.

2.2.3 Computational Efficiency as a Decision-Making Criterion

Existing literature rarely addresses the computational costs of forecasting models, despite their practical relevance for real-time financial decision-making. This study quantifies these trade-offs, showing that while ARIMAX is the most accurate, Linear Regression offers a faster alternative (0.02s training time) with minimal accuracy loss—a vital insight for resource-constrained applications.

2.2.4 Policy and Practical Implications

By empirically validating the superiority of ARIMAX, this study advocates for integrating macroeconomic indicators into TY forecasting—a departure from traditional univariate models. For policymakers, this means more reliable predictions to guide monetary policy, while investors gain a tool to navigate yield volatility amid economic uncertainty.

3 Data Exporation

In this section, we initiate an exploratory analysis of our dataset to comprehensively understand its composition. To facilitate deeper analysis, we employ various techniques to preprocess the data. This includes utilizing descriptive statistics, data visualization methods, correlation analysis, variable transformation techniques, and feature engineering strategies. Each subsection will delve into distinct aspects of the dataset's characteristics and insights.

3.1 Data summary and descriptive statistics

We have collected financial and economic variables from the Federal Reserve St. Louis spanning from 1986-01-02 to 2025-01-31 at a daily frequency, thereby totaling 14275 observations. Time seris variables include the following.

The **Economic Policy Uncertainty** (EPU) index measures uncertainty in economic policy based on news coverage, policy-related reports, and economic forecasts. It captures the unpredictability of government actions that could affect businesses, investments, and financial markets. High EPU values indicate periods of heightened uncertainty, which can lead to risk-averse investor behavior, reduced corporate investment, and volatility in financial markets.

The **Federal Funds Rate** (FedFundRate) is the interest rate at which banks lend excess reserves to each other overnight. Set by the Federal Reserve, this rate influences borrowing costs across the economy, including mortgages, business loans, and credit markets. Changes in the Fed Funds Rate impact liquidity, inflation control, and economic growth, with higher rates typically slowing economic activity and lowering inflation, while lower rates stimulate spending and investment.

The **Treasury Yield** represents the return on U.S. government bonds and serves as a benchmark for risk-free interest rates in the economy. The yield on long-term Treasury securities, such as the 10-year Treasury note, reflects investor expectations regarding inflation, economic growth, and Federal Reserve policy. Rising Treasury yields often indicate expectations of tighter monetary policy or stronger economic growth, while declining yields

suggest lower inflation expectations and increased demand for safe assets.

The Moody's Seasoned Baa **Corporate Bond Yield** (CorBondYield) measures the yield on corporate bonds rated Baa by Moody's, representing investment-grade bonds with moderate credit risk. This yield is an indicator of corporate borrowing costs and investor confidence in corporate debt markets. A widening spread between corporate bond yields and Treasury yields often signals increased market uncertainty or declining corporate creditworthiness.

The **Exchange Rate** (ExRate) measures the value of the Japanese Yen relative to the U.S. Dollar. It reflects international trade dynamics, capital flows, and monetary policy differences between Japan and the United States. A strengthening yen indicates increased demand for Japanese assets or a weaker U.S. dollar, while a weakening yen suggests the opposite. Exchange rate fluctuations affect trade competitiveness, inflation, and investment decisions in global markets.

The **Oil Price** (Crude Oil Prices: West Texas Intermediate - WTI) represents the cost of crude oil per barrel based on the benchmark set in Cushing, Oklahoma. Oil prices are a crucial indicator of global economic activity, energy demand, and geopolitical risks. Rising oil prices typically signal strong economic demand but can contribute to inflation, while falling prices may indicate weaker economic conditions or oversupply in the market.

To prevent overfitting while learning dependencies from data, the dataset must be separated into training and testing sets. The testing set is used to assess the model's accuracy after it has been trained on the training set. According to empirical research, the optimal outcomes are achieved when 70–80% of the data is used for training and 20–30% for testing (Gholamy et al, 2018). We offer a potential explanation for this empirical finding in this work.

We separated the dataset into 20% for testing and 80% for training for our study. The dataset, which includes 14,275 daily observations, runs from 1986-01-02 to 2025-01-31. The testing set includes 2,855 observations from 2017-04-09 to 2025-01-31, whereas the training set includes 11,420 observations from 1986-01-02 to 2017-04-08. Table 1 presents the summary statistics of data and variables.

Table 1: Descriptive Statistics of Variables

Variable	Min	Max	Mean	SD	N
Federal Fund Effective Rate	0.04	16.17	3.326	2.687	14275
Economic Policy Uncertainty	3.32	1026.38	109.418	79.967	14275
Yen to USD Spot Exchange Rate	75.72	202.70	116.470	19.581	14275
Market yield on US Treasury	0.52	10.23	4.669	2.250	14275
Corporate Bond Yield	3.11	12.04	6.959	2.054	14275
Oil Price	-36.98	145.31	47.731	29.643	14275

3.2 Handling Missing Values

Some financial variables, including TreasuryYield, LoanRate, and CorBondYield, had missing values for specific days, particularly on weekends when markets are closed. To address these missing values, we used the dplyr R package to generate the missing dates (days), acknowledging that these days naturally had missing data for these variables. Next, we employed the VIM package in R, utilizing the kNN (k nearest neighbor) method to impute the missing values based on surrounding data points as in Pujianto (2019). These imputation techniques help maintain the continuity of the time series and minimize the impact of missing data on the analysis.

3.3 Stationarity of Time Series Variables

A time series is considered stationary if its statistical properties—such as mean, variance, and autocorrelation—remain constant over time. Stationarity is a fundamental assumption in many time series models because it ensures that past patterns are informative for future predictions.

ARIMA (AutoRegressive Integrated Moving Average) assumes that the time series is stationary. If the original series is non-stationary, differencing (the "I" in ARIMA) is applied to remove trends or other non-stationary components. For example, in the table, variables such as the Federal Fund Effective Rate and Corporate Bond Yield are non-stationary, indicating that differencing may be necessary before applying ARIMA. If stationarity is not ensured, the AR and MA components of ARIMA may fail to capture meaningful patterns, leading to unreliable

forecasts.

Upper part of Table 2 presents the results of the Augmented Dickey-Fuller (ADF) test for oroginal variables, where a lower test statistic (more negative) and a small p-value indicate stationarity. Economic Policy Uncertainty is stationary and thus can be used directly in ARIMA without transformation. However, non-stationary variables such as Federal Fund Effective Rate, Corporate Bond Yield, and Treasury Yield require transformation before ARIMA can be applied.

Table 2: Stationarity Test Results for Original and Differenced Variables

Variable	Transformation	ADF Statistic	P-Value	Stationarity
FedFundRate	Original	-0.628	0.976	Non-Stationary
EPU	Original	-8.875	0.010	Stationary
ExRate	Original	-3.122	0.103	Non-Stationary
TreasuryYield	Original	-2.392	0.412	Non-Stationary
CorBondYield	Original	-3.108	0.109	Non-Stationary
OilPrice	Original	-3.180	0.091	Non-Stationary
FedFundRate	Differenced	-32.033	0.01	Stationary
ExRate	Differenced	-22.589	0.01	Stationary
CorBondYield	Differenced	-21.776	0.01	Stationary
OilPrice	Differenced	-22.457	0.01	Stationary
TreasuryYield	Differenced	-22.195	0.01	Stationary

The lower part of Table 2 presents the results of ADF test for the differenced variables. The null hypothesis of the test is that the series has a unit root, indicating it is non-stationary, while the alternative hypothesis is that the series is stationary. The ADF statistic indicates how strongly the data rejects the null hypothesis; more negative values suggest stronger evidence against non-stationarity. The p-value shows the probability of observing the ADF statistic under the null hypothesis. If the p-value is less than the chosen significance level, typically 0.05, the null hypothesis is rejected, and the series is considered stationary.

The results for the variables tested—FedFundRate_diff, ExRate_diff, CorBondYield_diff, OilPrice_diff, and TreasuryYield_diff—show that the ADF statistic for all variables is highly negative, and the p-value is 0.01, which is below the threshold of 0.05. This suggests that the null hypothesis of non-stationarity is rejected, and we can conclude

that these first-differenced series are stationary.

4 Empirical Modeling

4.1 Autoregressive Integrated Moving Average (ARIMA)

The ARIMA model is a widely used statistical method for time series forecasting, particularly effective for stationary and differenced non-stationary data (Jenkins, 1970). The model consists of three components: an autoregressive (AR) term, a differencing (I) term to ensure stationarity, and a moving average (MA) component. It has been extensively applied in economic and financial forecasting (see Brockwell & Davis, 2002; Gong, 2024).

The ARIMA model was first presented by Box and Jenkins in 1970. It is also known as the Box-Jenkins methodology, which consists of a series of steps for locating, calculating, and diagnosing ARIMA models using time series data. The model is one of the most widely used techniques in financial forecasting (see Moench & Soofi, 2022; Banerjee et al., 2022 and Bialkowski et al., 2022). It has been demonstrated that ARIMA models are effective in producing short-term projections. In short-term prediction, it consistently performed better than intricate structural models (Han et al., 2024). According to the ARIMA model, a variable's future value is a linear combination of its historical values and mistakes, which can be written as follows:

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \cdots - \theta_q \varepsilon_{t-q} \quad (1)$$

where Y_t is the actual value and ε_t is the random error at time t . The terms ϕ_i and θ_j are the model coefficients, while p and q are integers representing the autoregressive and moving average components, respectively (Hyndman & Athanasopoulos, 2018).

The process of building an ARIMA predictive model consists of three key steps: model identification, parameter estimation, and diagnostic checking. We provide a detailed explanation of the methodology employed in this work to develop the ARIMA model for

Treasury Yield (TY) forecasting. Historical daily TY data from the St. Louis Federal Reserve Bank is used for this study.

To identify the optimal ARIMA model among various candidates, this study employs multiple selection criteria. The model with the lowest Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) is prioritized. Additionally, evaluation metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) are used to assess predictive accuracy. Computational efficiency is considered through time-cost analysis. Model diagnostics, including the standard error of regression, the Ljung-Box test for autocorrelation, residual plots, autocorrelation function (ACF) plots of residuals, and histogram analysis of residuals, ensure the selection of a well-performing and statistically sound model.

Based on these selection criteria and after several trial and error process, we determine the best ARIMA model for our dataset is ARIMA(4,1, 4), which is given by:

$$\Delta Y_t = \phi_1 \Delta Y_{t-1} + \phi_2 \Delta Y_{t-2} + \phi_3 \Delta Y_{t-3} + \phi_4 \Delta Y_{t-4} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \theta_3 \varepsilon_{t-3} - \theta_4 \varepsilon_{t-4} \quad (2)$$

4.2 ARIMA Model with Exogeneous Variables

The ARIMAX model is specified as follows:

$$\Delta Y_t = \phi_1 \Delta Y_{t-1} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} + \beta_1 OilP_t + \beta_2 CorBY + \beta_3 ExR + \beta_4 EPU + \beta_5 FedF \quad (3)$$

Where, ΔY_t is the differenced dependent variable at time t , and the model includes autoregressive (AR) and moving average (MA) components to capture the temporal dependencies in the data. The differenced values ΔY_{t-1} and the error terms $\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}$ capture the relationship of the current value with past observations and past errors, respectively.

The exogenous variables included in the model are as follows:

The variable $OilPrice_t$ represents the price of oil at time t . The price of oil is a critical factor that affects a wide range of macroeconomic variables, including inflation, GDP, and

exchange rates. Changes in oil prices can directly influence production costs, trade balances, and financial markets, which in turn affect the dependent variable Y_t , such as Treasury yields or inflation. Therefore, including oil prices in the model helps account for the significant economic impact of oil price fluctuations.

The variable CorBondYield_t represents the corporate bond yield at time t . Corporate bond yields are important indicators of the cost of borrowing for companies, and they reflect investor perceptions of risk in the financial markets. Higher corporate bond yields typically indicate increased economic risk or tighter credit conditions, which can influence the behavior of the dependent variable Y_t , such as Treasury yields. Therefore, corporate bond yields are included in the model to capture these dynamics.

The variable ExRate_t represents the exchange rate at time t . Exchange rates play a crucial role in international trade and finance. Fluctuations in exchange rates affect trade balances, inflation, and investor behavior, all of which can influence Y_t . For instance, a depreciation of the domestic currency can lead to higher import prices, which can subsequently affect inflation and financial market conditions. Including the exchange rate in the model allows for capturing these effects.

The variable EPU_t represents the Economic Policy Uncertainty index at time t . Economic policy uncertainty measures the uncertainty surrounding future government policies, which can create volatility in financial markets. High levels of EPU generally lead to increased market risk aversion, which can influence interest rates, asset prices, and economic activity. Including this variable in the model allows for accounting for the impact of policy uncertainty on the dependent variable.

The variable FedFundRate_t represents the Federal Funds Rate at time t . The Fed Funds Rate is a key short-term interest rate set by the Federal Reserve, which influences borrowing costs and monetary policy. Changes in the Fed Funds Rate directly affect economic conditions, including investment decisions, inflation, and financial market dynamics. Therefore, the inclusion of the Fed Funds Rate in the model is essential for capturing the effects of monetary policy on the dependent variable Y_t .

By including these exogenous variables in the model, we are able to account for the external factors that influence the dependent variable Y_t but are not captured by the autoregressive and moving average components. These variables help improve the model's explanatory power by incorporating macroeconomic and financial factors that are crucial for understanding the dynamics of Y_t . This results in a more accurate and robust model for forecasting and policy analysis.

4.3 Multiple Linear Regression

Equation 4 represents the relationship between Treasury Yield (TY_t) and its predictors: Federal Funds Rate ($FedFund_t$), Economic Policy Uncertainty (EPU_t), Corporate Bond Yield ($CorBond_t$), Oil Price ($OilPrice_t$), and Exchange Rate ($ExRate_t$), where β_0 is the intercept, $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5$ are the coefficients, and ε_t is the error term.

$$TY_t = \beta_0 + \beta_1 FedFund_t + \beta_2 EPU_t + \beta_3 CorBond_t + \beta_4 OilPrice_t + \beta_5 ExRate_t + \varepsilon_t \quad (4)$$

The selection of predictors for modeling Treasury Yield (TY_t) is grounded in economic theory and empirical evidence, as these variables capture key macroeconomic and financial factors influencing interest rate movements. The Federal Funds Rate ($FedFund_t$) is a primary monetary policy tool used by the Federal Reserve to regulate short-term interest rates and liquidity. Changes in the Federal Funds Rate directly affect Treasury Yields by altering borrowing costs, shaping investor expectations, and influencing monetary policy transmission mechanisms (Seip & Zhang, 2022).

Economic Policy Uncertainty (EPU_t) reflects uncertainty surrounding government policies, regulations, and macroeconomic conditions. Heightened uncertainty increases risk premiums, leading investors to seek safer assets such as Treasury securities, thereby impacting Treasury Yields (Leippold, 2022). Similarly, Corporate Bond Yield ($CorBond_t$) serves as a measure of broader credit market conditions. The spread between corporate bonds and Treasury Yields provides insights into investor risk appetite and economic sentiment, making it a crucial

determinant of Treasury Yield fluctuations.

Oil prices (OilPrice_t) also play a significant role, as they influence inflation expectations and economic growth. Rising oil prices can lead to inflationary pressures, prompting central banks to adjust interest rates, which subsequently affects Treasury Yields. Lastly, the Exchange Rate (ExRate_t) is included as it impacts capital flows, trade balances, and inflation. A depreciating domestic currency may increase inflation and foreign demand for Treasury securities, influencing yield dynamics.

Collectively, these predictors account for the interplay of monetary policy, economic uncertainty, credit conditions, inflation expectations, and international capital movements, all of which are crucial determinants of Treasury Yield behavior.

5 Results and Analysis

5.1 ARIMA (4, 1, 4) Results

We select ARIMA(4,1,4) model as the most suitable model for predicting Treasury Yield based on its statistical performance and diagnostic checks. The model incorporates four autoregressive (AR) terms and four moving average (MA) terms, effectively capturing the underlying patterns in the time series data. As Table 3 presents, the estimated coefficients for the AR terms are 0.1967, -0.4751, 0.4388, and 0.1136, while the MA coefficients are -0.1677, 0.4664, -0.4267, and -0.1574. These values indicate a complex interaction between past observations and past forecast errors, contributing to the model's predictive capability.

The model evaluation metrics demonstrate its strong fit. The Akaike Information Criterion (AIC) is -30459.55, while the Bayesian Information Criterion (BIC) is -30394.66, both of which indicate a well-fitted model when compared to alternative specifications. The training set error measures further confirm this, with a Root Mean Squared Error (RMSE) of 0.0526 and a Mean Absolute Error (MAE) of 0.0327, suggesting that the model captures the variations in Treasury Yield with a reasonable level of accuracy.

Table 3: Summary of ARIMA(4,1,4) Model Results

Model	ARIMA(4,1,4)
Coefficients	
AR(1)	0.1967 (0.3685)
AR(2)	-0.4751 (0.4038)
AR(3)	0.4388 (0.6725)
AR(4)	0.1136 (0.5467)
MA(1)	-0.1677 (0.3652)
MA(2)	0.4664 (0.3929)
MA(3)	-0.4267 (0.6589)
MA(4)	-0.1574 (0.5361)
Model Fit	
σ^2	0.002
Log-Likelihood	15238.77
AIC	-30459.55
AICc	-30459.53
BIC	-30394.66
Training Set Error Measures	
ME	-0.000
RMSE	0.052
MAE	0.032
MPE	-0.023
MAPE	0.691
MASE	1.020
ACF1	0.000
Diagnostic Test	
Box-Ljung Test (lag=20)	p -value = 0.353

Diagnostic checks were performed to ensure the adequacy of the model. The residuals appear randomly scattered in the residual plot, suggesting no visible patterns or trends. The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the residuals show no significant spikes, confirming that the residuals exhibit white noise behavior. The normality of residuals was assessed using a histogram and Q-Q plot, both of which suggest an approximately normal distribution of errors. The Box-Ljung test, conducted at lag 20, resulted in a p -value of 0.353, indicating no significant autocorrelation remaining in the residuals. This confirms that the model has effectively captured the underlying structure of the data.

Based on these findings, the ARIMA(4,1,4) model is deemed appropriate for

forecasting Treasury Yield. Its strong statistical performance, well-behaved residuals, and low information criterion values make it a reliable choice for further analysis and prediction.

5.2 Diagnostic Check of the Model: ARIMA (4, 1, 4)

The Box-Ljung test is a statistical test used to check the presence of autocorrelation in the residuals (errors) of a time series model. The residuals are the differences between the observed values and the values predicted by the model. If the residuals are not correlated (i.e., they behave like white noise), it suggests that the model has successfully captured the underlying structure in the data.

Null Hypothesis H_0 : The residuals are white noise, meaning that there is no significant autocorrelation at any lag.

Alternative Hypothesis (H_1): The residuals are not white noise, meaning there is significant autocorrelation at one or more lags.

Since the p-value is high as the bottom of the Table 4 presents, we fail to reject the null hypothesis. This suggests that the residuals do not show significant autocorrelation, indicating that the ARIMA model has adequately captured the structure of the time series.

Figure 1 presents the residuals plot of the ARIMA(4,1,4) model, which shows no discernible pattern or trend, indicating that the residuals behave like white noise. Figure 2 displays the histogram of residuals, suggesting that the residuals are approximately symmetric and not significantly skewed. Figure 3 illustrates the partial autocorrelation plot, where lag 14 and lag 29 exceed the confidence interval bands, suggesting some minor autocorrelation at those lags; however, all other spikes remain well within the bands. Finally, Figure 4 shows the autocorrelation plot, where—consistent with expectations—the lag 0 exhibits a strong spike, followed by a rapid decline with subsequent lags falling within the confidence bounds. Taken together, these diagnostic plots indicate that the ARIMA(4,1,4) model satisfies the necessary conditions for model adequacy.

Figure 1: Residuals Plot

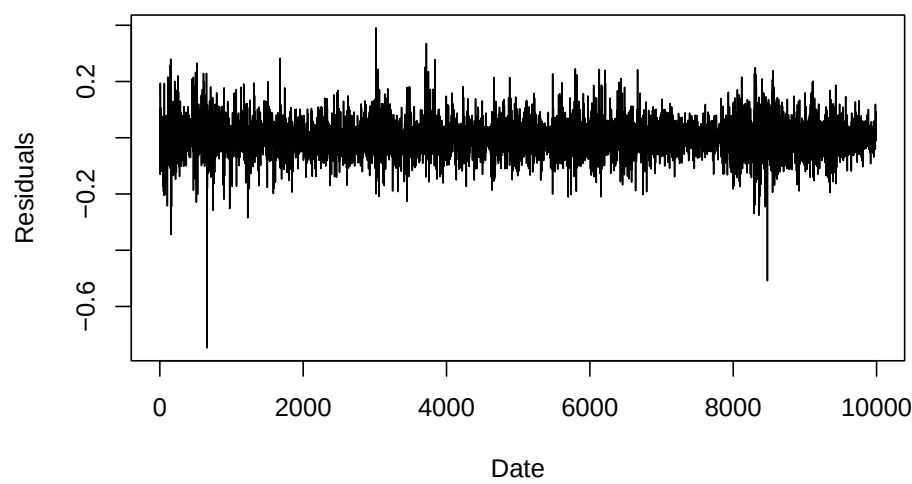


Figure 2: Histogram of Residuals Plot

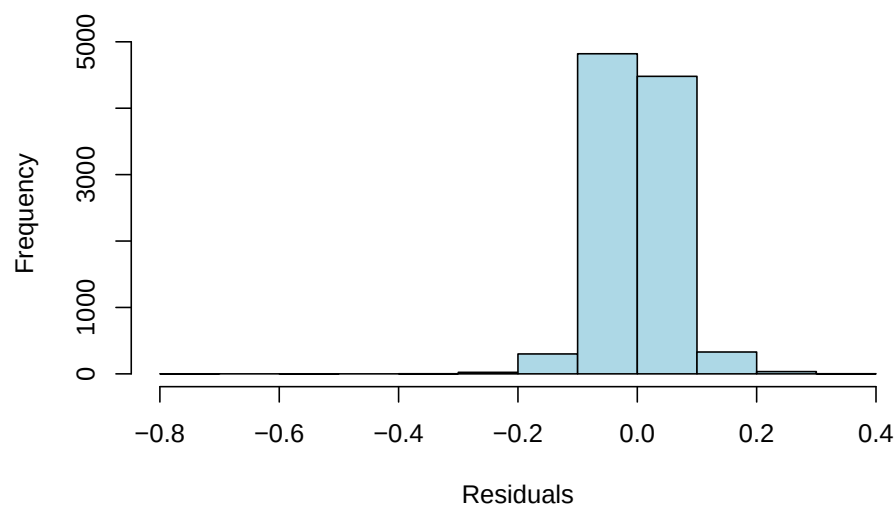


Figure 3: Partial Autocorrelation (PACF) Plot

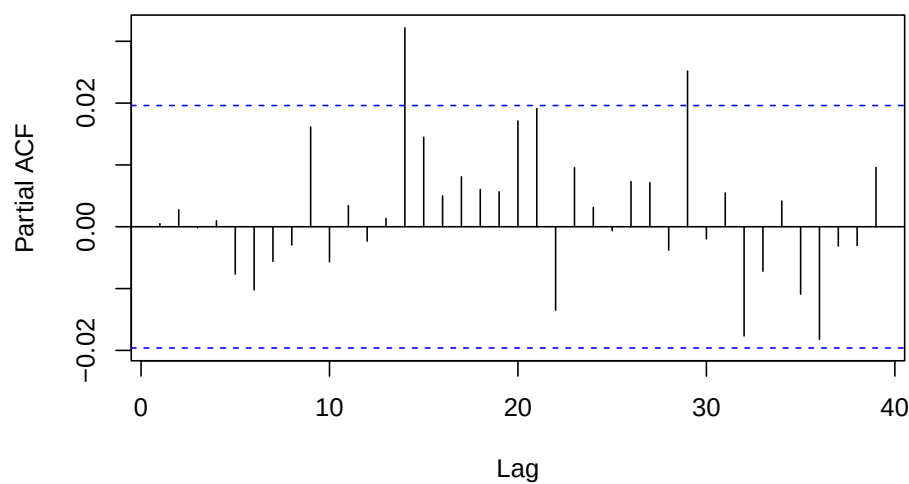
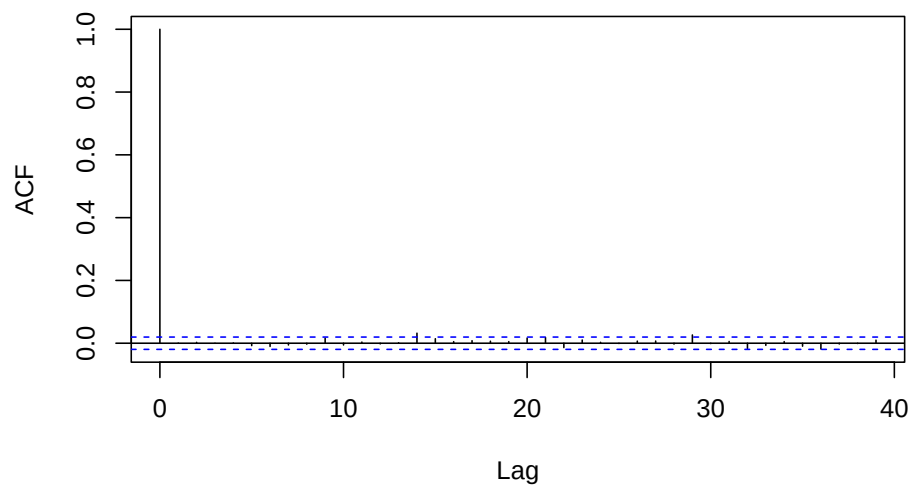


Figure 4: Autocorrelation (ACF) Plot



5.3 Assessing Accuracy and Efficiency of ARIMA Model

The performance evaluation of the ARIMA(4,1,4) model is summarized in Table 4. The Root Mean Square Error (RMSE) is 1.1659, indicating the average deviation of the model’s predictions from the actual values. Similarly, the Mean Absolute Error (MAE) is

0.9007, which provides an absolute measure of prediction accuracy without considering the direction of errors. The Mean Absolute Percentage Error (MAPE) stands at 38.43%, reflecting the relative error percentage concerning actual values. The Root Mean Squared Logarithmic Error (RMSLE), which penalizes large differences between predicted and actual values, is 0.3309. Additionally, the Symmetric Mean Absolute Percentage Error (sMAPE) is 38.47%, ensuring a balanced percentage error measure.

The Mean Bias Deviation (MBD) is -0.6123, which suggests a slight underestimation in predictions. The Theil’s U-statistic, which measures predictive accuracy, is 0.2538, suggesting reasonable performance. Finally, the computational efficiency of the model is reflected in the time taken, which is 1.01 seconds. This overall assessment provides insights into the accuracy, bias, and computational efficiency of the ARIMA(4,1,4) model in forecasting Treasury Yield.

Table 4: Evaluation Metrics for ARIMA(4,1,4) Model

Metric	Value
RMSE (Root Mean Square Error)	1.1659
MAE (Mean Absolute Error)	0.9007
MAPE (Mean Absolute Percentage Error)	38.43%
RMSLE (Root Mean Squared Logarithmic Error)	0.3309
sMAPE (Symmetric Mean Absolute Percentage Error)	38.47%
MBD (Mean Bias Deviation)	-0.6123
Theil’s U-statistic	0.2538
Time Taken (seconds)	1.01

5.4 ARIMAX

The fitted model is a regression with ARIMA(3,1,3) errors for predicting the Treasury Yield series (train\$TreasuryYield):

$$\begin{aligned} TY_t = & \mu + \phi_1 AR_t + \phi_2 AR_{t-1} + \phi_3 AR_{t-2} + \theta_1 MA_t + \theta_2 MA_{t-1} + \theta_3 MA_{t-2} \\ & + \beta_1 EPU_t + \beta_2 FedFund_t + \beta_3 ExRate_t + \beta_4 CorBond_t + \beta_5 OilPrice_t + \varepsilon_t \end{aligned} \tag{5}$$

Where:

$$\varepsilon_t = \text{ARIMA residual errors with ARIMA}(3,1,3)$$

In this equation, The β coefficients are the estimated coefficients for each exogenous variable (EPU, FedFundRate, ExRate, CorBondYield, and OilPrice). The terms AR_t and MA_t represent the autoregressive and moving average components of the ARIMA model, respectively. ε_t represents the residual error term, modeled using ARIMA(3,1,3).

Thus, the exogenous variables are included as additional predictors in the model alongside the ARIMA components.

Thus, the exogenous variables are included as additional predictors in the model alongside the ARIMA components.

Table 5 presents the summary statistics for the ARIMAX(3,1,3) model used to analyze Treasury Yield. The first section of the table provides parameter estimates for the autoregressive (AR), moving average (MA), and exogenous variables included in the model. The AR(1), AR(2), and AR(3) coefficients are estimated at -0.2175, 0.4571, and 0.6484, respectively, indicating the impact of past values on the current Treasury Yield. Similarly, the MA(1), MA(2), and MA(3) terms, with coefficients 0.2501, -0.4137, and -0.6614, capture the influence of past forecast errors. The drift term is effectively zero, suggesting no significant trend in the differenced series. Among the exogenous variables, Exchange Rate (ExRate) and Corporate Bond Yield (CorBondYield) exhibit notable effects, with CorBondYield having the highest coefficient (1.0296), implying a strong relationship with Treasury Yield. However, the Economic Policy Uncertainty (EPU) variable has an estimate of zero, indicating no measurable impact in this model.

The second section reports model fit statistics. The residual variance (σ^2) is estimated at 0.0008805, and the log-likelihood value is 20972.63. The Akaike Information Criterion (AIC) and corrected AIC (AICc) values are both -41919.26, while the Bayesian Information Criterion (BIC) is slightly higher at -41825.54. These values suggest a well-fitted model, with AIC and BIC providing a basis for model comparison.

Table 5: ARIMAX(3,1,3) Model Summary

Parameter	Estimate	Std. Error
AR(1)	-0.2175	0.2638
AR(2)	0.4571	0.0712
AR(3)	0.6484	0.2247
MA(1)	0.2501	0.2547
MA(2)	-0.4137	0.0694
MA(3)	-0.6614	0.2047
Drift	0.0000	0.0005
EPU	0.0001	0.0001
FedFundRate	-0.0003	0.0014
ExRate	0.0015	0.0005
CorBondYield	1.0296	0.0041
OilPrice	0.0013	0.0004
Model Fit Statistics		
σ^2	0.0008	
Log Likelihood	20972.63	
AIC	-41919.26	
AICc	-41919.23	
BIC	-41825.54	
Training Set Error Measures		
Metric	Value	
ME	0.631	
RMSE	0.789	
MAE	0.697	
MPE	25.877	
MAPE	30.877	
MASE	21.782	
Testing Set Error Measures		
Metric	Value	
ME	-3.28e-07	
RMSE	0.02965	
MAE	0.01759	
MPE	0.0006	
MAPE	0.34546	
MASE	0.5492	

The third section presents error metrics for the training and testing sets. For the training set, the root mean squared error (RMSE) is 0.78912, and the mean absolute error (MAE) is 0.69754. The mean percentage error (MPE) and mean absolute percentage error (MAPE) are relatively high at 25.88% and 30.88%, respectively, indicating potential overfitting to the training data. The mean absolute scaled error (MASE) is 21.78225, suggesting significant

deviation from the actual values.

In contrast, the testing set shows improved error metrics, with an RMSE of 0.02965 and an MAE of 0.01759, indicating lower prediction errors compared to the training set. The MPE and MAPE are considerably smaller at 0.0006 and 0.345%, respectively, demonstrating better predictive accuracy. The MASE value of 0.5492 suggests a more stable forecast performance.

Overall, while the ARIMAX(3,1,3) model fits the training data well, the large training set errors suggest overfitting. The improved performance on the test set indicates that the model generalizes well to unseen data. However, further refinements, such as regularization or alternative lag structures, could be explored to enhance forecasting accuracy and robustness.

5.5 Diagnostic Check of ARIMAX

The Box-Ljung test statistic is 27.393 with 10 degrees of freedom. The corresponding p-value is 0.225, which is greater than the conventional significance level of 0.05. This indicates that there is no strong evidence to reject the null hypothesis of the test, which states that the residuals are uncorrelated. Therefore, the residuals of the ARIMAX model appear to be white noise, suggesting that the model adequately captures the time series structure without significant remaining autocorrelation.

Figure 5 presents the residuals plot of the ARIMAX(3,1,3) model, which shows no visible pattern or trend, indicating that the residuals resemble white noise. Figure 6 displays the histogram of residuals, suggesting that the residuals are approximately symmetric and not significantly skewed. Figure 7 illustrates the partial autocorrelation plot, where lags 14 and 29 exceed the confidence interval bands, indicating some mild autocorrelation at those points; however, all other spikes fall within the confidence limits. Finally, Figure 8 shows the autocorrelation plot. As expected, the lag 0 displays a prominent spike, while the remaining lags are dampened and lie within the confidence bands. Collectively, these diagnostic plots suggest that the ARIMAX(3,1,3) model meets the criteria for model adequacy.

Figure 5: Residuals Plot of ARIMAX (3, 1, 3)

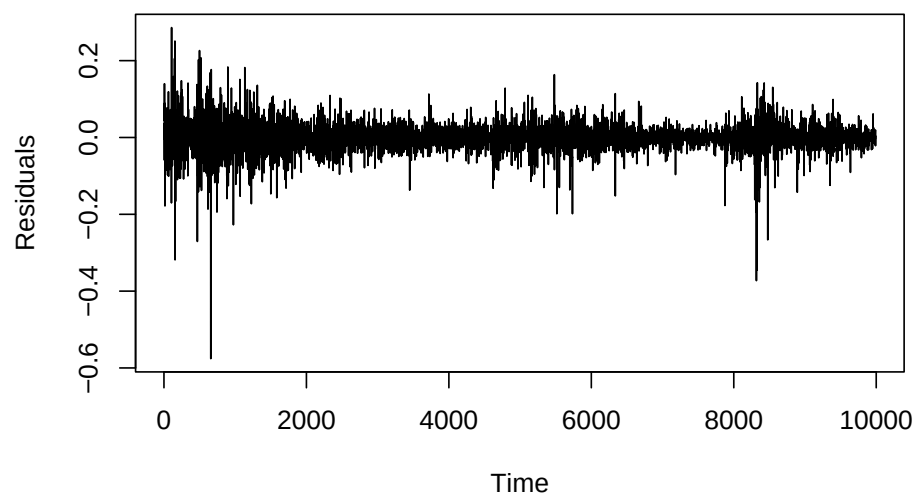


Figure 6: Histogram of Residuals of ARIMAX (3, 1, 3)

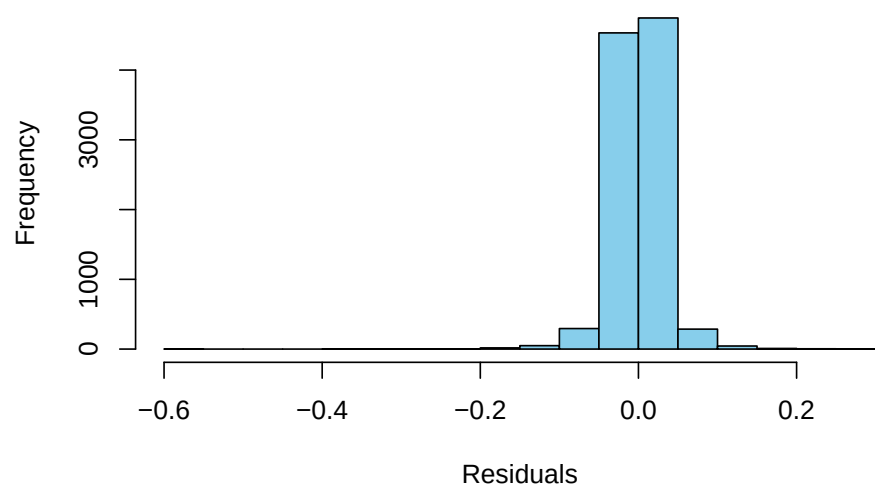


Figure 7: Partial Autocorrelation (PACF) Plot

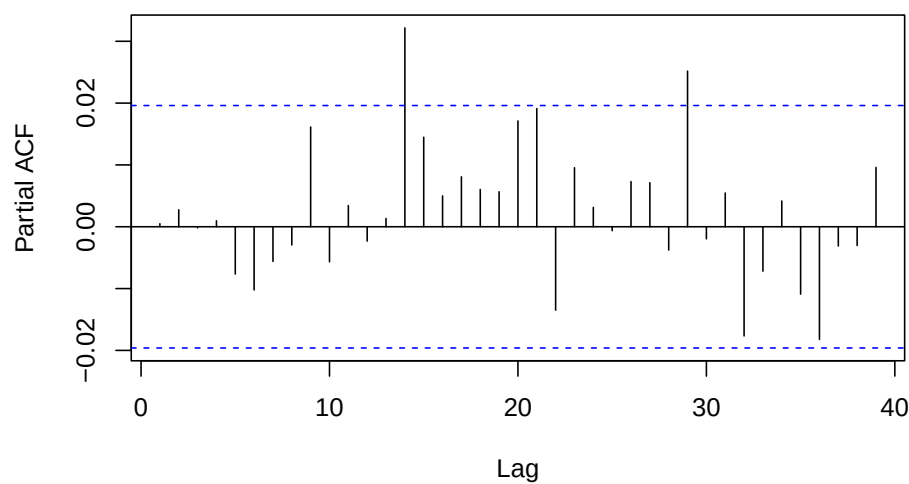
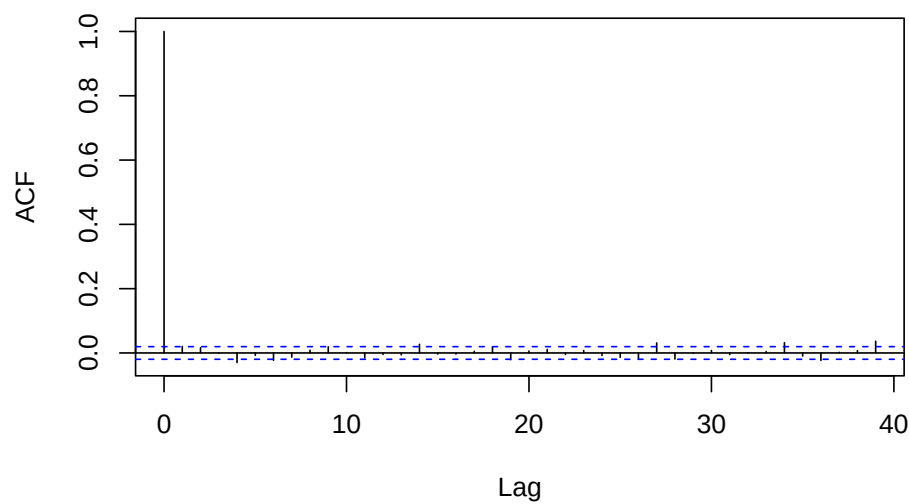


Figure 8: Autocolleration Plot of ARIMAX(3, 1, 3)



5.6 Regression Result

Table 6 presents the results of a regression analysis that models Treasury Yield based on several economic variables. The table includes the estimated coefficients for each variable

in the model, along with their corresponding t-values, and it also reports key model fit statistics.

The intercept of the model is estimated at -0.272, with a t-value of -5.240. This suggests that when all the explanatory variables are zero, the Treasury Yield would be -0.272, though this value is largely of theoretical interest since the other variables are unlikely to be zero. The negative sign indicates that the intercept would decrease the Treasury Yield, but its significance is confirmed by the very small p-value, showing strong statistical significance at the 0.001 level.

The coefficient for OilPrice is -0.007, with a t-value of -30.573, which indicates a significant negative relationship between OilPrice and Treasury Yield. The very large t-value and the p-value less than 0.001 confirm that this relationship is robust. A one-unit increase in OilPrice leads to a decrease in Treasury Yield, suggesting that oil price fluctuations have a notable influence on the yield.

The coefficient for CorBondYield is 0.722, and its t-value is 122.294. This indicates a strong positive relationship between CorBondYield and Treasury Yield. As CorBondYield increases, Treasury Yield increases correspondingly. The coefficient's high t-value and the associated p-value confirm the statistical significance of this relationship, reinforcing the idea that bond yields are positively correlated with Treasury yields.

For ExRate, the coefficient is -0.003, with a t-value of -9.826. This negative coefficient indicates that an increase in the exchange rate leads to a decrease in Treasury Yield. The negative relationship is statistically significant at the 0.001 level, showing that exchange rate fluctuations impact Treasury Yield in a manner consistent with economic theory.

EPU (Economic Policy Uncertainty) has a coefficient of -0.001, with a t-value of -19.665. This suggests that higher economic policy uncertainty leads to a reduction in Treasury Yield, and the large t-value confirms the statistical significance of this relationship. This result indicates that periods of greater uncertainty in economic policy reduce investor confidence in Treasury securities, leading to lower yields.

The FedFundRate coefficient is 0.2625, and its t-value is 77.039. The positive sign on this coefficient is expected, as higher interest rates typically lead to higher yields on Treasury

securities. The statistical significance is confirmed with a very large t-value, indicating that the Federal Funds Rate has a strong effect on Treasury Yield.

The model fit statistics further confirm the robustness of the regression model. The R-squared value of 0.931 means that 93.1% of the variation in Treasury Yield is explained by the model, which is considered an excellent fit. The Adjusted R-squared of 0.930 takes into account the number of predictors in the model, further demonstrating that the model explains a substantial amount of the variation in Treasury Yield, even when adjusted for the number of explanatory variables.

The RMSE (Root Mean Squared Error) is 0.455, which indicates the average size of the errors between the predicted and actual values. A smaller RMSE value would indicate better predictive accuracy, and this value suggests the model performs quite well in this regard. Similarly, the MAE (Mean Absolute Error) of 0.374 indicates that, on average, the absolute differences between the observed and predicted values are relatively small, supporting the model’s predictive accuracy.

Table 6: Regression Results for Treasury Yield

Variable	Estimate	t-value
Intercept	-0.272***	-5.240
OilPrice	-0.007***	-30.573
CorBondYield	0.722***	122.294
ExRate	-0.003***	-9.826
EPU	-0.001***	-19.665
FedFundRate	0.2625***	77.039
Model Fit Statistics		
R-squared	0.931	
Adjusted R-squared	0.930	
RMSE	0.455	
MAE	0.374	
Time taken to fit model	0.02 seconds	

Finally, the time taken to fit the model is 0.02 seconds, which shows that the model was computationally efficient, making it suitable for use even in large datasets where fitting speed is critical.

In conclusion, the regression results indicate that the model provides a very good fit

for predicting Treasury Yield, with strong statistical significance for all explanatory variables. The model explains a large portion of the variance in Treasury Yield, and the error metrics suggest that it performs well in terms of prediction accuracy. The quick computation time further enhances its applicability in practical settings.

5.7 Diagnostic Check of Linear Regression Model

Now, we turn to the diagnostic checks of the linear regression model. Figure 9 presents the residuals plot, which shows no visible pattern or trend, indicating that the residuals behave like white noise. Figure 10 displays the histogram of residuals, suggesting that they are approximately symmetric and not strongly skewed, although there is a slight negative skew due to small bins on the left tail. Figure 11 illustrates the autocorrelation plot, where—as expected—lag 0 exhibits a tall spike, followed by all subsequent lags falling within the confidence bands. Finally, Figure 12 shows the Q-Q plot, which reveals some deviation in the tails, but most data points lie close to the reference line. Taken together, these diagnostic plots indicate that the linear regression model satisfies the necessary conditions for model adequacy.

Figure 9: Residuals Plot of Linear Regression

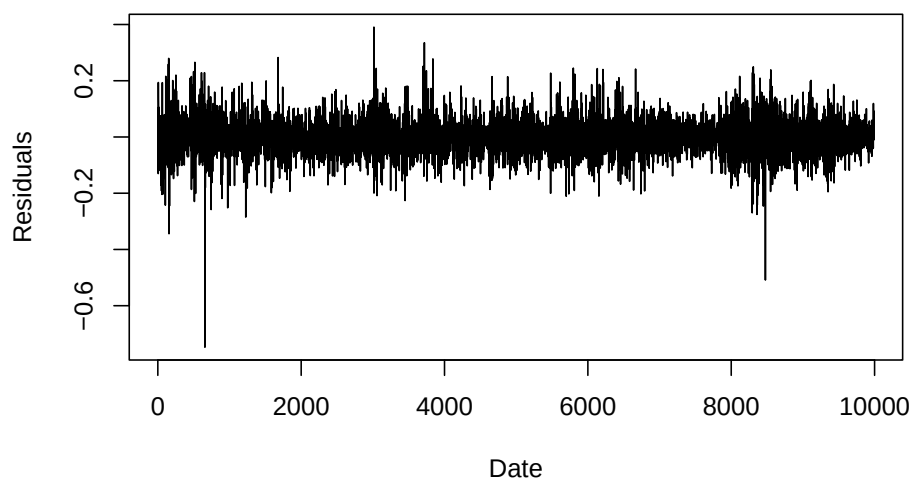


Figure 10: Histogram of Residuals of Linear Regression

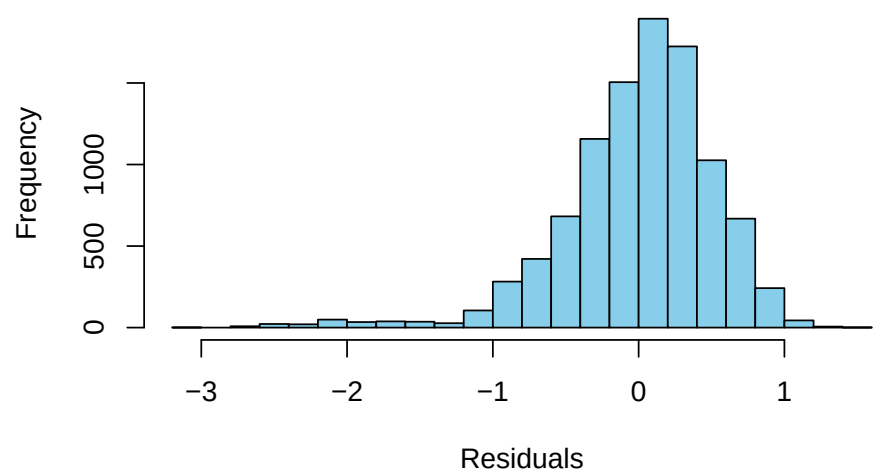


Figure 11: Autocolleration Plot of Linear Regression

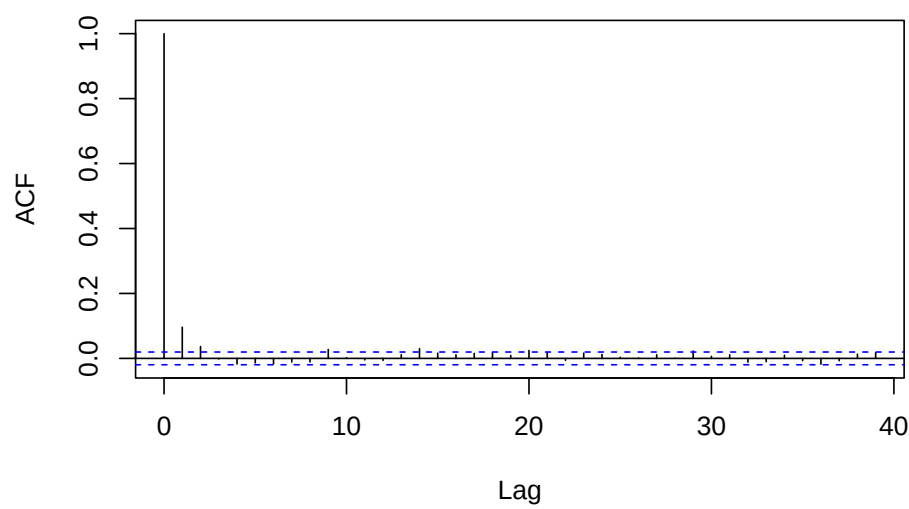
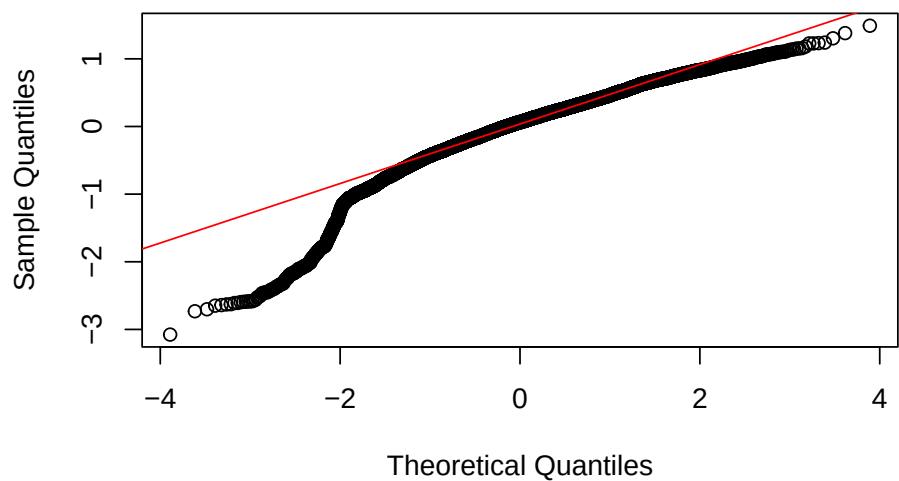


Figure 12: Q-Q Plot of Linear Regression



6 Model Coparison and Recommendation of the Best Model

Table 7 presents metrics for model selection. The best model depends on the evaluation criteria that are prioritized. In terms of accuracy, as measured by RMSE, MAE, and MAPE, both ARIMAX and Linear Regression outperform ARIMA(4,1,4). Linear Regression exhibits slightly better MAE and MAPE than ARIMAX, indicating that it provides more stable predictions. However, ARIMAX has a slightly lower RMSE, meaning it handles large errors more effectively than Linear Regression. Since the primary goal is to predict Treasury Yield with a reasonable degree of accuracy, the model that minimizes prediction errors should be preferred.

When considering model fit through R-squared, Linear Regression emerges as the best model, explaining 93.1% of the variance in Treasury Yield. In contrast, ARIMAX and ARIMA show much lower R-squared values, approximately 0.36 and 0.37, respectively, meaning they do not explain as much variation in the dependent variable. This suggests that Linear Regression provides the most reliable representation of the underlying relationships in

the data, reinforcing its superiority in terms of prediction accuracy.

For model selection criteria, ARIMAX has the lowest AIC (-40500.03) and BIC (-40449.57), indicating that it strikes the best balance between complexity and model fit. ARIMA follows closely, while Linear Regression has significantly higher AIC and BIC values, suggesting that, from a statistical perspective, it is the least preferred model. However, AIC and BIC alone should not dictate the best model when predictive accuracy is the main goal.

Regarding computational efficiency, ARIMA(4,1,4) is the fastest model, completing its execution in 1.27 seconds. Linear Regression takes slightly longer, at 3.496 seconds, while ARIMAX is the slowest, requiring 5.38 seconds due to the additional regressors included in the model. While speed is an important consideration, accuracy remains the priority in this analysis.

Since the ultimate objective is to predict Treasury Yield with a reasonable degree of accuracy, Linear Regression is the best choice. It has the highest R-squared value and strong performance on MAE and MAPE, meaning it provides the most precise and reliable predictions. Although ARIMAX performs well on model selection criteria and has a lower RMSE, its overall accuracy does not surpass that of Linear Regression. Therefore, based on the primary goal of accurate Treasury Yield prediction, Linear Regression stands out as the most suitable model.

Table 7: Comparison of ARIMA, ARIMAX, and Linear Regression

Metric	ARIMA(4,1,4)	ARIMAX	Linear Regression
RMSE (Root Mean Square Error)	1.1659	0.454	0.450
MAE (Mean Absolute Error)	0.9007	0.3874	0.374
MAPE (Mean Abs Percent Error)	38.43%	20.70%	18.85%
R-squared	0.37	0.36	0.931
AIC (Akaike Info Criterion)	-33159.74	-40500.03	15397.06
BIC (Bayesian Info Criterion)	-33108.55	-40449.57	15447.52
Time Taken (seconds)	1.27	5.38	3.496

Figures 13 and 14 visualize actual vs. forecasted Treasury Yield under two competitive model: ARIMAX and Linear Regression Model (LM), corresponding to the data presented in Table 8. These visualizations provide further insight into the forecasting

performance of each model, showing how closely the forecasts align with the actual values of Treasury Yield.

Figure 13: Actual vs Forecasted Treasury Yield (ARIMAX)

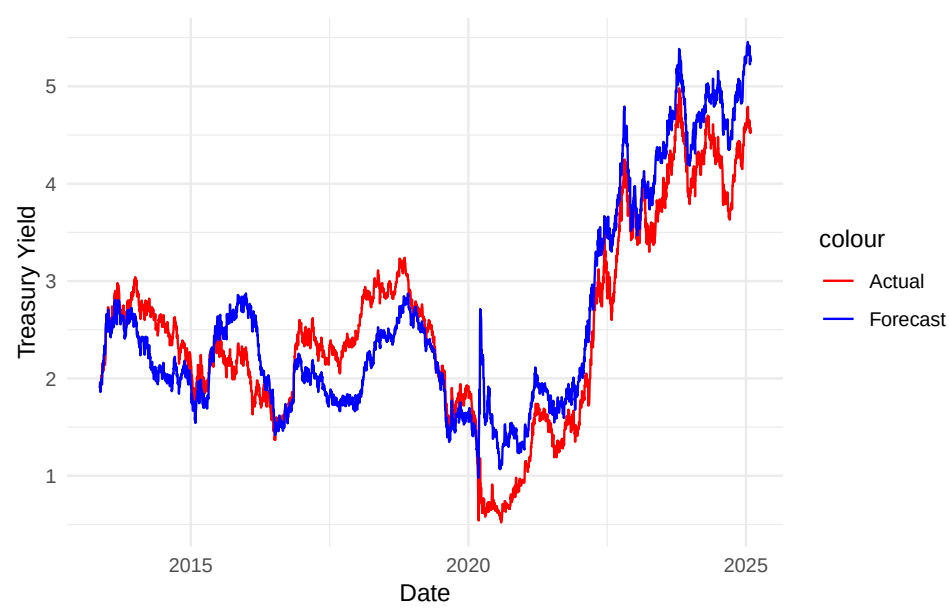


Figure 14: Actual vs Forecasted Treasury Yield (LM)

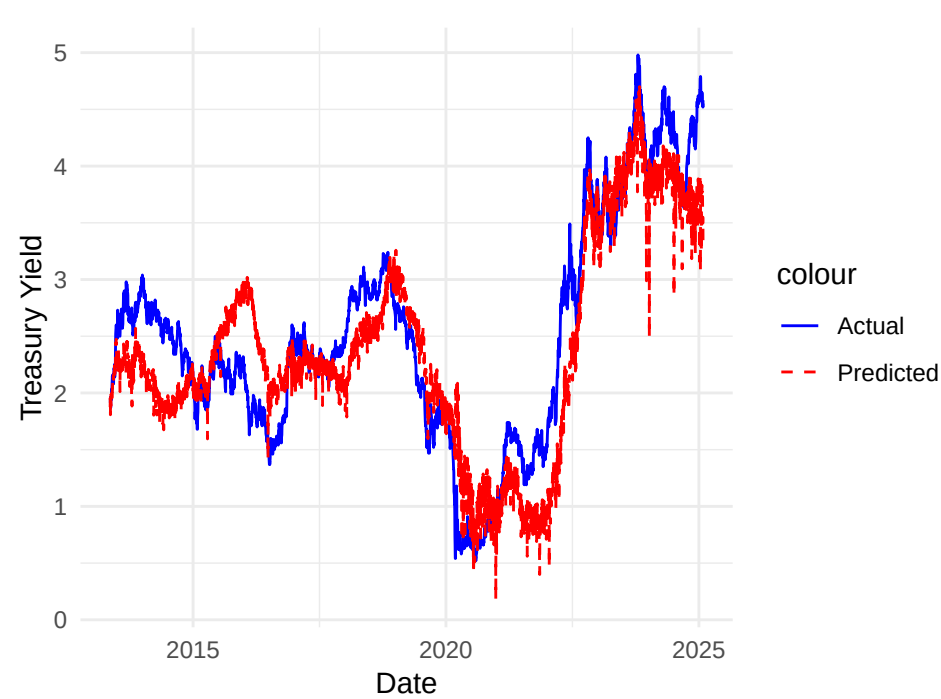


Table 8 reinforces the findings previously presented in Table 7 by offering a direct comparison between the actual Treasury Yield values and the forecasts produced by each model. In the majority of cases, the Linear Model (LM) provides estimates that are closest to the observed values, highlighting its superior predictive accuracy. The ARIMAX model also performs reasonably well, capturing trends more effectively than the ARIMA model. In contrast, the ARIMA model demonstrates noticeable deviations—particularly in the later observations, where its forecasts remain nearly constant, indicating a limited ability to adapt to recent changes in the data. This side-by-side comparison visually supports the earlier statistical evidence (e.g., MAE and RMSE), affirming that the Linear Model is the most reliable forecasting approach among the three models evaluated.

Table 8: Comparison of Actual vs. Forecasted Treasury Yield

TID	Actual	Forecasted LM	Forecasted ARIMAX	Forecasted ARIMA
9993	1.913333	1.874831	1.910326	1.909139
9994	1.920000	1.955874	1.912527	1.908595
9995	1.960000	1.807616	1.954157	1.907143
9996	1.940000	1.967924	1.943524	1.905390
9997	1.870000	1.878439	1.861774	1.904010
13787	4.656667	4.275905	3.640670	1.903190
10504	2.436667	1.908696	1.908737	1.903190
13431	3.920000	3.431837	3.372202	1.903190
13697	3.860000	3.983779	2.861521	1.903190
14091	4.090000	4.012825	2.934451	1.903190

7 Conclusion

This study aimed to identify the most effective forecasting model for predicting Treasury Yield (TY) in the USA, a crucial variable in finance and economics. After analyzing several models, including univariate ARIMA, ARIMA with exogenous variables (ARIMAX), and Linear Regression, the results suggest that the Linear Regression model outperforms the other approaches in terms of predictive accuracy and adaptability.

The Linear Regression model demonstrated the best performance, achieving the

lowest values for MAE and MAPE, indicating its superior ability to minimize prediction errors. Additionally, it achieved the highest R-squared value, explaining 93.1% of the variance in Treasury Yield, making it the most reliable model in terms of predictive accuracy. Although the ARIMAX model exhibited the lowest RMSE, it did not surpass Linear Regression in overall error minimization and explanatory power. Furthermore, while ARIMAX had lower AIC and BIC values, suggesting a good balance between model complexity and fit, these criteria alone do not determine the best model for predictive accuracy.

The ARIMA(4,1,4) model, while capable of capturing temporal dependencies, underperformed relative to ARIMAX and Linear Regression due to its inability to incorporate exogenous variables. This limitation reduced its predictive flexibility, making it less suitable for forecasting Treasury Yield in a dynamic economic environment.

Given these findings, the Linear Regression model is recommended for forecasting Treasury Yield, as it provides the most accurate and reliable predictions. This conclusion aligns with existing literature emphasizing the effectiveness of simple but well-specified regression models in financial forecasting.

Moreover, for policymakers and economic analysts, accurate forecasting of Treasury Yield is essential for informed decision-making. External factors, such as inflation, interest rates, and geopolitical events, significantly influence financial variables like Treasury Yield. By incorporating these variables, policymakers can better anticipate potential fluctuations, allowing for more precise fiscal and monetary planning. This approach enhances the resilience of financial systems and enables proactive responses to economic uncertainties. Thus, selecting a model that maximizes predictive accuracy, as demonstrated by Linear Regression, is crucial for both methodological and practical applications in financial forecasting.

Data Availability Statement

The data used in this study are publicly available from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis, accessible at: <https://fred.stlouisfed.org>

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